f4184fc596403b9d638783cf57adfe4c75c605f6356fbc91338530e9831e9e16.
f4184fc596403b9d638783cf57adfe4c75c605f6356fbc91338530e9831e9e16.

How do we reach consensus on who owns what money without a central authority?

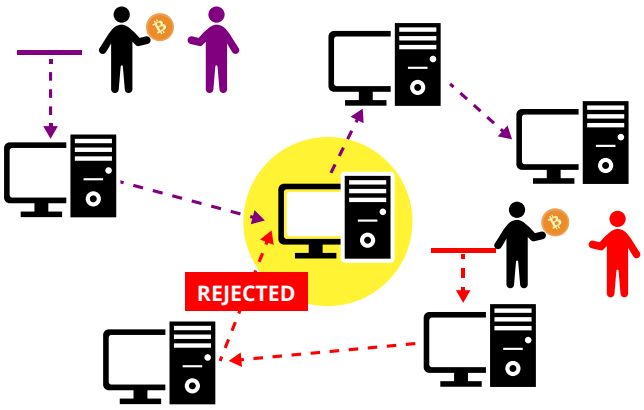
Hashing and code. Let's explain.

Imagine you have 1 ***bitcoin*** and you want to send it to your friend as a birthday gift. You send the ***bitcoin*** to your friend's address, but then you realize that you owe your ex-boyfriend money and should have sent the ***bitcoin*** to him instead. In a moment of panic, you decide to be sneaky and create a new transaction to send the same 1 ***bitcoin*** to your ex-boyfriend. This is what we call a "double spend."

But wait, how does the network prevent this from happening? It's simple. The nodes on the network detect conflicting transactions and only allow one to go through based on a set of rules known as the "**consensus rules**." In this case, it's likely that the transaction to your ex-boyfriend would be rejected since it was sent after the original transaction to your friend. However, it's just a matter of luck which transaction is picked up first by a miner.

Thanks to the *blockchain*, **everyone on the network is able to agree on the current state of the ledger**. This helps to prevent **double-spending** and fraud, making it a secure and trustworthy system, just like a digital version of the "honor system." So, next time you send ***bitcoin***, you can relax knowing that the network has got your back.

So how does ***Bitcoin*** actually solve this? Well, let's find out.

- ***Bitcoin*** prevents double-spending by implementing a confirmation mechanism and maintaining a universal ledger (*blockchain*).
- Transactions are added to the *blockchain* in a chronologically-ordered, time-stamped manner.
- To prevent double spending, only the first transaction to receive enough confirmations (usually 6) is included in the *blockchain*, while the others are discarded.
- Transactions on the *blockchain* are irreversible and impossible to tamper with.

Bitcoin Mining Node Rejects Transaction from Mempool